

1 查看网络配置:

```
2  
3 cmd overlay lookup --verbose com.android.connectivity.resources  
com.android.connectivity.resources:array/config_ethernet_interfaces  
4 cmd overlay lookup --verbose com.android.connectivity.resources  
com.android.connectivity.resources:string/config_ethernet_iface_regex  
5 dumpsys ethernet
```

6
7
8 在Android系统中配置网络的是: CarConnectivityOverlayMtk

9
10 单android的系统, 非hyper的系统, CarConnectivityOverlayMtk需要删除, 不要参与编译, 不然会导致需要指定ETH节点才能ping通

```
11 device/mediatek/vendor/common/car/device.mk  
12 46 # Runtime Resource Overlay for Connectivity  
13 47 PRODUCT_PACKAGES += \  
14 48 CarConnectivityOverlayMtk
```

```
15  
16  
17 auto8676p1_64_bsp:/ # cmd overlay lookup --verbose  
com.android.connectivity.resources  
com.android.connectivity.resources:array/config_ethernet_interfaces  
    <  
18 Resolution for 0x06010000  
19 For config - zh-rCN-ldltr-sw960dp-w1707dp-h788dp-xlarge-long-notround-  
nowidecg-lowdr-land-car-night-180dpi-finger-keysoft-nokeys-nonav-1920x972-v34  
20 Found initial: /vendor/overlay/CarConnectivityOverlayMtk.apk #10  
21 Overlaid: /vendor/overlay/CarConnectivityOverlayMtk.apk #10  
22 Best matching is from default configuration of  
com.mediatek.connectivity.resources.car.overlay  
23 eth0;11,12,13,14,15,28;ip=192.168.1.100/24 gateway=192.168.1.1
```

24
25 CarConnectivityOverlayMtk.apk代码位置是:

```
26 device/generic/car/common/overlay_packages/ConnectivityOverlay/
```

27
28 单Android系统使用的网络配置文件是:

```
29 :  
vendor/mediatek/proprietary/packages/overlay/vendor/ConnectivityOverlay/res/values/config  
.xml
```

30 双系统Android端使用的网络配置应用文件是:

```

31 vendor/mediatek/proprietary/packages/overlay/vendor/ConnectivityOverlay/res_auto/values/c
onfig.xml
32
33 eth0;11,12,13,14,15,28说明:
34 1、eth0接口的说明可以查看:
packages/modules/Connectivity/service/ServiceConnectivityResources/res/values/config.xml
文件中的说明
35 2、11,12,13,14,15,28这些数字是 Network Capabilities、可以参考
packages/modules/Connectivity/framework/src/android/net/NetworkCapabilities.java文件中的说
明
36 注意: 添加dns需要添加11和28
37 vendor/mediatek/proprietary/packages/overlay/vendor/ConnectivityOverlay/res_auto/values/c
onfig.xml目录
38 <string-array translatable="false" name="config_ethernet_interfaces">
39 <!-- for adb, no internet -->
40 <item>eth0;11,13,14;ip=192.168.1.2/24 gateway=192.168.1.1</item>
41 <!-- for data share, internet, Override Transport Cellular transport -->
42 <item>eth1;11,12,13,14,15,28;ip=172.17.0.2/24 gateway=172.17.0.1
dns=8.8.8.8;0</item>
43 <!-- for xx -->
44 <item>eth2;11,12,13,14,15,28;ip=172.17.1.2/24 gateway=172.17.1.1</item>
45 <!-- for xx -->
46 <item>eth3;11,12,13,14,15,28;ip=172.17.2.2/24 gateway=172.17.2.1</item>
47 <!-- for xx -->
48 <item>eth4;11.12.13.14.15.28;ip=172.17.3.2/24 gatewav=172.17.3.1</item>
49 <!-- for xx -->
50 <item>eth5;11,12,13,14;ip=172.20.1.44/24 gatewav=172.20.1.1</item>
51 </string-array>
52
53 配置ip挂载上自动配置:
54 packages/modules/Connectivity/service-t/src/com/android/server/ethernet/
55
56 diff --git a/service-t/src/com/android/server/ethernet/EthernetTracker.java b/service-
t/src/com/android/server/ethernet/EthernetTracker.java
57 index 2603d4a6e4..a50164a51f 100644
58 --- a/service-t/src/com/android/server/ethernet/EthernetTracker.java
59 +++ b/service-t/src/com/android/server/ethernet/EthernetTracker.java
60 @@ -164,6 +164,21 @@ public class EthernetTracker {
61         OsConstants.NETLINK_ROUTE, NetlinkConstants.RTMGRP_LINK);
62     }
63
64 +    // M
65 +    public static boolean isVirtioNetInterface(String iface) {
66 +        if (iface == null || iface.isEmpty())

```

```

67 +         return false;
68 +     try {
69 +         String path = android.system.Os.readlink("/sys/class/net/" + iface +
70 +             "/device/driver");
71 +         Log.i(TAG, iface + "->" + path);
72 +         if (path != null && path.contains("virtio_net"))
73 +             return true;
74 +     } catch (Exception e) {
75 +         Log.e(TAG, "Error readlink " + iface, e);
76 +     }
77 +     return false;
78 + }
79
80 private void onNewLink(String ifname, boolean linkUp) {
81     if (!mFactory.hasInterface(ifname) && !ifname.equals(mTetheringInterface)) {
82         Log.i(TAG, "onInterfaceAdded, iface: " + ifname);
83     }
84     Log.i(TAG, "interfaceLinkStateChanged, iface: " + ifname + ", up: " +
85         linkUp);
86     updateInterfaceState(ifname, linkUp);
87     // M after interface up
88     if (ifname != null && (ifname.contains("eth0") || ifname.contains("eth5"))
89         && isVirtioNetInterface(ifname)) {
90         try {
91             IpConfiguration ipConfig = getIpConfiguration(ifname);
92             if (ipConfig != null && ipConfig.getStaticIpConfiguration() != null
93                 && ipConfig.getStaticIpConfiguration().getIpAddress() !=
94                 null) {
95                 LinkAddress linkAddress =
96                     ipConfig.getStaticIpConfiguration().getIpAddress();
97                 mNetd.networkAddInterface(INetd.LOCAL_NET_ID, ifname);
98                 mNetd.interfaceAddAddress(ifname,
99                     linkAddress.getAddress().getHostAddress(),
100                     linkAddress.getPrefixLength());
101                 List<android.net.RouteInfo> routes = new ArrayList<>();
102                 routes.add(new android.net.RouteInfo(linkAddress, null,
103                     ifname));
104                 NetdUtils.addRoutesToLocalNetwork(mNetd, ifname, routes);
105             }
106         } catch (ServiceSpecificException | RemoteException e) {
107             Log.e(TAG, "Failed to add " + ifname + " to local table: ", e);
108         }
109     }
110 }

```

```
105     }
106
107
108 2、yocto侧对以太网的管理，可以用yocto自带的网络管理模块NetworManager进行:代码路径: meta/meta-
openembedded/meta-networking/recipes-connectivity/networkmanager
109 将平台用以太网线连接到路由器的LAN口，此模块会透过dhcp获得ip，然后设置路由等，实现联网。
110
111 commit 79007533f53f6fff2250e9190d49ef05c5cb85f5 (HEAD -> pvt_single_dev)
112 Author: shenjiyun <jyshen@pvetec.com>
113 Date: Wed Nov 6 14:20:26 2024 +0800
114
115 feat(Net):配置pcie以太网enp1s0默认ip为192.168.1.100
116
117 Description: 配置pcie以太网enp1s0默认ip为192.168.1.100
118
119 Product: MT8676
120 Project: P39
121 Build: apass
122 Change-Id: Ie6c79bfb4ccb95ab35cf3ddd27b1039899e537c8
123
124 diff --git a/meta-networking/recipes-
connectivity/networkmanager/files/enp1s0.nmconnection b/meta-networking/recipes-
connectivity/networkmanager/files/enp1s0.nmconnection
125 new file mode 100755
126 index 0000000..a6e4da1
127 --- /dev/null
128 +++ b/meta-networking/recipes-connectivity/networkmanager/files/enp1s0.nmconnection
129 # 添加文件，配置默认ip和节点
130 @@ -0,0 +1,12 @@
131 +[connection]
132 +id=enp1s0
133 +type=ethernet
134 +interface-name=enp1s0 # 指定网络节点
135 +
136 +[ipv4]
137 +method=manual //配置ip
138 +addresses1=192.168.1.100/24
139 +gateway=192.168.1.1
140 +
141 +[ipv6]
142 +method=ignore
143 \ No newline at end of file
144
```

```

145 diff --git a/meta-networking/recipes-
connectivity/networkmanager/networkmanager_1.46.0.bb b/meta-networking/recipes-
connectivity/networkmanager/networkmanager_1.46.0.bb
146 index 8184fcf..6a53edc 100644
147 --- a/meta-networking/recipes-connectivity/networkmanager/networkmanager_1.46.0.bb
148 +++ b/meta-networking/recipes-connectivity/networkmanager/networkmanager_1.46.0.bb
149 @@ -39,11 +39,13 @@ SRC_URI = " \
150     file://${BPN}.initd \
151     file://enable-dhpcpd.conf \
152     file://enable-iwd.conf \
153 +   file://enp1s0.nmconnection \ #添加文件拷贝到Networkmanager中
154
155     SRC_URI:append:libc-musl = "${@bb.utils.contains('DISTRO_FEATURES', 'ld-is-lld', '
file://0001-linker-scripts-Do-not-export-_IO_stdin_used.patch', '', d)}"
156
157     SRC_URI[sha256sum] = "722649e25362693b334371473802a729b0ec9ee283375096905f868808e74068"
158
159     S = "${WORKDIR}/NetworkManager-${PV}"
160
161     # ['auto', 'symlink', 'file', 'netconfig', 'resolvconf']
162 @@ -320,4 +322,15 @@ do_install:append() {
163     if ${@bb.utils.contains('PACKAGECONFIG', 'dhcpcd', 'true', 'false', d)}; then
164         install -Dm 0644 ${WORKDIR}/enable-dhpcpd.conf
165         ${D}${nonarch_libdir}/NetworkManager/conf.d/enable-dhpcpd.conf
166     fi
167 +   # 安装自定义文件到正确目录、必须要放在NetworkManager/system-connections/下才会自动配置、
168     # 执行生效
169 +   install -d ${D}${sysconfdir}/NetworkManager/system-connections/
170 +   install -m 0600 ${WORKDIR}/enp1s0.nmconnection
171     ${D}${sysconfdir}/NetworkManager/system-connections/
172 +
173 }
174
175 +# 重启服务
176 +pkg_postinst_${PN}() {
177 +   if [ -n "$D" ]; then
178 +       exit 1
179 +   fi
180 +   systemctl restart NetworkManager || true
181 +}
182
183 \ No newline at end of file
184
185 3. 在yocto下另起服务可以参照:
186
187 yocto侧起服务可以参考这个服务meta/meta-mediatek-mt8676-hyp/recipes-apps/uos-init/files/uos-
init.service,

```

180 然后在meta/meta-mediatek-mt8676-hyp/recipes-auto/images/mtk-core-image-auto8676-uos.bb加上
该服务既可

181

二、lua文件关于网络相关配置の説明：

```
1  --  vmnet
2  --[[
3  void setVmnet(std::string id, std::string ip, std::string netmask)
4  void setVmnet(std::string id, std::string ip, std::string netmask, std::string fe_MAC)
5  uos_config:setVmnet("sample_eth0_if", "172.16.16.1", "255.255.255.0",
6  "02:22:ee:b9:55:66");
7  --]]
8  uos_config:setVmnet("vmnet0", "192.168.2.1", "255.255.255.0", "02:22:ee:b9:55:66")
9  uos_config:setVmnet("vmnet-vccmni0", "172.17.0.1", "255.255.255.0", "02:22:ee:b9:55:68")
10 uos_config:setVmnet("vmnet-vccmni1", "172.17.1.1", "255.255.255.0")
11 uos_config:setVmnet("vmnet-vccmni2", "172.17.2.1", "255.255.255.0")
12 uos_config:setVmnet("vmnet-vccmni3", "172.17.3.1", "255.255.255.0")
13 _ _*****
14 上述中std::string fe_MAC字段修改的事Android端的mac地址，yocto端的mac地址是随机的，如果要修改的
15 话可以参照上面NetworkManager修改ip地址的修改，增加
16 [ethernet]
17 cloned-mac-address=00:11:22:33:44:55 # 固定 MAC
18 # 或使用随机化策略
19 cloned-mac-address=random # 每次连接随机生成
20 cloned-mac-address=stable # 为每个网络生成固定随机地址
```